

Computational Neuroscience: Analysis of brain patterns using machine learning algorithms

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Abstract

This Bachelor's Thesis explores the application of machine learning algorithms to the analysis of perceptual decision-making data in order to identify behavioral patterns associated with sensory evidence accumulation and potential confirmation bias. The study is based on experimental data obtained from a task in which participants repeatedly categorize sequences of visual stimuli. The project combines methods from computational neuroscience and machine learning to evaluate whether participants' decisions can be predicted from experimental variables such as sensory evidence, previous responses and reaction times. In addition, the work aims to examine the relative contribution of these variables to model predictions and to assess whether machine learning can serve as a complementary tool to traditional computational approaches such as the Drift Diffusion Model.

Keywords - Computational neuroscience, perceptual decision-making, machine learning, confirmation bias, sensory evidence accumulation, behavioral analysis.

1. Problem definition and motivation

Decision-making is a fundamental process in human cognition that involves the progressive integration of sensory information in order to reach a conclusion or judgment. Several studies in the field of cognitive neuroscience have shown that this process is not entirely objective, but can be influenced by prior decisions or internal beliefs of the individual. One of the most widely studied phenomena in this context is confirmation bias, which describes the tendency of individuals to give greater weight to information that confirms their previous decisions or beliefs while minimizing the influence of contradictory information.

In the context of perceptual decision-making, this bias may emerge during the process through which the brain gradually integrates sensory information over time in order to reach a decision. This process, known as sensory evidence accumulation, consists of combining different samples of perceptual information until sufficient evidence is available to make a decision. Several recent studies have suggested that prior decisions may influence how this new information is integrated. In some cases, this effect may cause information that confirms the previous decision to be weighted more heavily than information that contradicts it. As a consequence, individuals may assign greater importance to sensory samples that are consistent with their previous response while reducing the influence of inconsistent information, which may promote the repetition of initial decisions even when new potentially contradictory evidence appears.

Computational models traditionally used to study these processes, such as the Drift Diffusion Model, allow researchers to mathematically describe the dynamics of evidence accumulation during decision-making. In this type of model, decision-making is conceptualized as a process in which sensory evidence is progressively accumulated until a certain threshold is reached, leading to the final decision. This framework makes it possible to analyze how different factors, such as the quality of the evidence or the influence of prior decisions, can affect both the speed and direction of the decision-making process. However, these models often require specific theoretical assumptions about the operation of the cognitive processes involved.

In recent years, machine learning algorithms have shown a strong ability to detect complex patterns in experimental data and to model human behavior without the need to impose strict prior theoretical structures. This potential opens the possibility of using these methods to analyze perceptual decision-making data and identify patterns associated with cognitive biases.

In this context, this TFG focuses on the application of machine learning algorithms to analyze data obtained from a perceptual decision-making experiment based on the repeated presentation of sequences of visual stimuli. The objective is to investigate to what extent these models can predict participants' decisions and detect patterns consistent with the presence of confirmation bias in the process of sensory evidence integration.

This approach allows the combination of methodologies from computational neuroscience with artificial intelligence techniques, with the aim of providing new tools for studying the cognitive mechanisms involved in human decision-making.

1.1. Relevance of the project

Understanding how humans integrate sensory information when making decisions is a central question in cognitive neuroscience. While traditional computational models such as the Drift Diffusion Model have provided valuable theoretical frameworks for describing these processes, recent advances in machine learning offer new opportunities for analyzing behavioral data from a data-driven perspective.

The relevance of this project lies in exploring the potential of machine learning algorithms as complementary tools for studying perceptual decision-making. By analyzing experimental data using predictive models, this work aims to identify behavioral patterns associated with the integration of sensory evidence and the influence of prior decisions.

In particular, the project aims to examine whether machine learning methods can detect patterns consistent with confirmation bias during the decision-making process. In this way, the study contributes both to the methodological development of computational neuroscience and to the broader understanding of cognitive biases in human cognition.

2. Context and state of the art

Perceptual decision-making has been widely studied in the fields of **cognitive neuroscience** and **experimental psychology**. These tasks require individuals to integrate sensory information over time until reaching a categorical decision, that is, a choice between

a limited set of possible categories. Traditionally, these processes have been modeled using computational evidence accumulation models such as the Drift Diffusion Model. This model describes decision-making as a process of gradual accumulation of sensory evidence until a decision threshold is reached.

Several studies have shown that perceptual decisions are not based only on the currently available sensory evidence, but can also be influenced by prior decisions or by the recent history of responses. This phenomenon is known as choice history bias and manifests as a systematic tendency to repeat previous decisions or to integrate new information in a manner consistent with prior choices.

One mechanism that may explain this behavior is confirmation bias, a cognitive bias through which individuals tend to prioritize information that confirms their previous beliefs or decisions while reducing the influence of contradictory information. In the context of perceptual decision-making, this bias may emerge during the process of sensory evidence accumulation, such that information samples consistent with the previous decision have a greater impact on the final decision than those that are inconsistent.

Recent studies have suggested that this type of bias may emerge from a selective overweighting of evidence consistent with the previous decision, causing information consistent with the prior decision to carry greater weight than contradictory information during the sensory integration process. This mechanism has been identified in several experimental paradigms and has been interpreted as a form of confirmation bias in perceptual decision-making.

Furthermore, recent research has highlighted the role of metacognition in these processes. Metacognition refers to the ability of individuals to evaluate the correctness of their own decisions, often through confidence judgments. Some studies have shown that the level of confidence in a decision can modulate how new information is integrated, in other words, decisions made with a high level of confidence tend to be more resistant to revision, whereas greater metacognitive efficiency may facilitate error detection and the correction of previous decisions.

In this context, recent experimental studies have investigated how individuals integrate sensory information when the same stimulus is presented repeatedly. These paradigms make it possible to analyze the extent to which prior decisions influence the interpretation of new sensory evidence. The results suggest that participants tend to repeat their initial decisions and assign greater weight to information samples that are consistent with their previous responses, reflecting a pattern of evidence integration similar to confirmation bias.

Although classical computational models have been widely used to describe these processes, in recent years there has been increasing interest in applying machine learning methods to the study of human behavior. These methods allow the identification of complex patterns in experimental data and can complement traditional cognitive models in the analysis of decision-making processes.

In this sense, the application of machine learning algorithms can provide new tools for analyzing perceptual decision-making data, enabling the prediction of participants' behavior and the detection of patterns associated with cognitive biases. This approach is particularly

relevant in the emerging field of computational neuroscience, which combines computational models, experimental data and artificial intelligence techniques to understand the neural and cognitive mechanisms underlying human behavior.

3. Objectives

The main objective of this TFG is to analyze the process of perceptual decision-making through the application of machine learning algorithms, with the aim of identifying patterns in participants' behavior and detecting potential cognitive biases in the integration of sensory evidence.

Specifically, this work is based on the analysis of data obtained from a perceptual decision-making experiment in which participants are required to categorize sequences of visual stimuli presented repeatedly. This experimental context makes it possible to study the extent to which prior decisions may influence how new sensory information is integrated, as well as to analyze the role of factors such as reaction time and consistency between consecutive decisions.

Based on this general framework, the following specific objectives are defined :

Objective 1. To perform an exploratory analysis of the experimental data in order to identify the variables relevant to the decision-making process, such as sensory evidence, participants' prior decisions and reaction times.

Objective 2. To develop and train machine learning models capable of predicting participants' decisions based on the available experimental variables.

Objective 3. To compare the performance of different machine learning algorithms in predicting participants' decisions in order to determine which models achieve the best predictive performance.

Objective 4. To analyze which variables have the greatest impact on the models' predictions, with the aim of identifying possible patterns consistent with the presence of confirmation bias in the integration of sensory evidence.

Objective 5. To compare the results obtained from the machine learning models with those reported in previous studies based on traditional computational models, in order to assess the extent to which machine learning models can capture similar behavioral patterns.

Objective 6. To evaluate the potential of machine learning algorithms as a complementary tool for the study of cognitive processes in the field of computational neuroscience.

4. Methodology and work plan

To achieve the objectives proposed in this thesis, a methodology based on the analysis of experimental data and the application of machine learning algorithms will be followed in order to model and predict participants' behavior in a perceptual decision-making task.

4.1. Experimental data

This work will be based on a dataset obtained from a perceptual decision-making experiment in which participants are required to evaluate sequences of visual stimuli and categorize them according to their mean orientation. In each experimental trial, participants make a categorical decision after observing a sequence of stimuli and the same sequence may be presented again in subsequent repetitions.

The dataset includes relevant information about participants' behavior during the task, such as the decision made in each presentation of the stimulus, the reaction time associated with each decision, the characteristics of the presented stimulus and the relationship between consecutive decisions within the same experimental trial.

These data make it possible to analyze how sensory information and prior decisions influence subsequent decision-making.

4.2. Data preprocessing and exploratory analysis

In the first phase of the project, a data preprocessing procedure will be carried out in order to prepare the data for use in machine learning models. This stage will include tasks such as:

- data cleaning and validation
- organization of the experimental variables
- generation of new derived variables (for example, consistency with prior decisions or information about stimulus repetitions)

In addition, an exploratory data analysis will be performed in order to identify initial patterns in participants' behavior and to better understand the relationships between the different variables in the dataset.

4.3. Development of machine learning models

Once the data have been prepared, several supervised machine learning models will be developed with the aim of predicting participants' decisions based on the available experimental variables.

Initially, relatively simple classification models will be implemented, such as logistic regression and support vector machines (SVM), as these methods provide interpretable baselines for binary classification problems. Subsequently, more flexible approaches based on tree-based learning methods may also be explored, such as Random Forest and Gradient Boosting, which are capable of capturing potentially non-linear relationships between variables.

To ensure robust evaluation of the models, the dataset will be divided into training and testing subsets, allowing the models to be trained on one portion of the data and evaluated on unseen observations. This procedure helps reduce the risk of overfitting and provides a more reliable estimate of model performance.

The purpose of using multiple models is to compare their ability to capture patterns in the data and to identify which experimental variables most strongly influence participants' decisions.

4.4. Analysis of results and detection of biases

Once the models have been trained, a detailed analysis of the obtained results will be carried out. Model performance will be evaluated using standard classification metrics such as accuracy, precision, recall and F1 score. When appropriate, additional metrics such as the area under the ROC curve (AUC) may also be considered in order to provide a more complete evaluation of predictive performance.

Beyond predictive accuracy, the analysis will focus on model interpretability by examining the contribution of the different variables used in the models. Particular attention will be given to variables related to participants' prior decisions, stimulus characteristics and reaction times.

This analysis will allow assessment of whether the predictive patterns identified by the models are consistent with the presence of confirmation bias in the process of sensory evidence integration.

5. Expected results

It is expected that the machine learning models developed in this project will be able to predict participants' decisions with performance above chance level, indicating that the experimental variables contain relevant information about the decision-making process.

In particular, variables related to prior decisions and stimulus characteristics are expected to have a significant influence on the models' predictions. If confirmation bias is present in the data, information consistent with previous decisions may have a stronger impact on the predicted outcomes than contradictory information.

Overall, the results are expected to provide insights into how prior decisions influence perceptual decision-making and to illustrate the potential of machine learning techniques for identifying behavioral patterns associated with cognitive biases.

6. Project planning

The development of this thesis will be divided into different phases that will allow the project objectives to be addressed progressively. These phases include the literature review, the preprocessing and analysis of the experimental data, the development of machine learning models and the analysis of the obtained results.

Phase 1: Literature review and detailed definition of the problem

During this first phase, a review of the scientific literature related to perceptual decision-making, evidence accumulation models and cognitive biases such as confirmation

bias will be conducted. Previous studies that use computational models and machine learning techniques in the analysis of human behavior will also be examined.

This stage will make it possible to establish the theoretical framework of the project and to define more precisely the objectives and methodological approach of the study.

Phase 2: Data preprocessing and exploratory analysis

In this phase, the dataset obtained from the experiment will be analyzed. Data cleaning, structuring and preparation tasks will be carried out in order to make the data suitable for use in machine learning models. An exploratory analysis will also be performed with the aim of identifying initial patterns in participants' behavior and better understanding the relationships between the available variables.

Phase 3: Development of machine learning models

Once the data has been prepared, different machine learning models will be developed in order to predict participants' decisions based on the available experimental variables. Several classification algorithms will be implemented and trained, including models such as logistic regression, support vector machines and tree based methods such as Random Forest or Gradient Boosting.

Phase 4: Model evaluation and analysis of results

In this phase, the performance of the developed models will be evaluated and compared in order to determine which algorithms achieve the best predictive performance. In addition, the importance of the different variables used in the predictions will be analyzed. This analysis will help identify possible patterns consistent with the presence of confirmation bias in the process of sensory evidence integration.

The results obtained from the machine learning models will also be compared with those reported in previous studies based on traditional computational models.

Phase 5: Writing and revision of the final thesis

Finally, the thesis document will be written, including the description of the theoretical context, the methodology used, the results obtained and the discussion of the project's conclusions.

7. Project time line - Gantt Chart

